University of Cambridge Mathematical Tripos Part III

Functional Analysis

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Course description

This course covers many of the major theorems of abstract Functional Analysis. It is intended to provide a foundation for several areas of pure and applied mathematics. We will cover as many of the following topics as time permits.

- Hahn–Banach theorems on the extension of linear functionals. Locally convex spaces. Discussion of the duals of the spaces $L_p(\mu)$ and $C(K)$.
- Weak and weak-star topologies. Hahn–Banach theorems on separation of convex sets. Theorems of Mazur, Goldstine and Banach–Alaoglu. Reflexivity and local reflexivity. Uniform convexity. The Milman–Pettis theorem. Duals of the spaces $L_p(\mu)$.
- Extreme points and the Krein–Milman theorem. Partial converse and the Banach–Stone theorem.
- Schauder bases in Banach spaces. Basis techniques and applications: compact and strictly singular operators; Pelczyński's decomposition method; the Eberlein–Šmulian theorem.
- The Bochner integral. The Krein–Šmulian theorem.
- Non-linear theory of Banach spaces. Lipschitz functions and the Lipschitz classification problem. Spaces with the Radon–Nikodym property. Linearization via differentiability and Rademacher's theorem. Linearization via invariant means. Uniform classification and Ribe's theorem.
- Banach algebras, elementary spectral theory. Commutative Banach algebras and the Gelfand representation theorem. Holomorphic functional calculus.
- Hilbert space operators, C^*-algebras. The Gelfand–Naimark theorem. Spectral theorem for commutative C^*-algebras. Spectral theorem and Borel functional calculus for normal operators.

Prerequisites

Thorough grounding in basic topology and analysis. Some knowledge of basic functional analysis and basic measure theory. In spectral theory we will make use of basic complex analysis. For example, Cauchy's theorem, Cauchy's integral formula and the maximum modulus principle. Much of the background material will be recalled either in lectures or via handouts.

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1 The Hahn–Banach extension theorems

1.1 Dual spaces

Let X be a (real or complex) normed space. Recall from II Linear Analysis that the *dual space* X^* is defined as the space of all bounded linear functions on X, which is a Banach space with respect to the operator norm

$$||f|| := \sup\{|f(x)| \colon x \in B_X\}$$

Above, we denote by B_X the closed unit ball in X

$$B_X := \{x \in X \colon ||x|| \leq 1\}$$

Similarly, we denote by S_X the unit sphere

$$S_X := \{x \in X \colon ||x|| = 1\}$$

Examples *Normed spaces and their dual*

- $\ell_p^* \cong \ell_q$ for $1 \leq p < \infty, 1 < q \leq \infty$ with $p^{-1} + q^{-1} = 1$
- $c_0^* \cong \ell_1$
- If $\mathcal{H}$ is a Hilbert space, then $\mathcal{H}^* \cong \mathcal{H}$ (conjugate-linear in the complex case)

Notational remark. Let X, Y be normed spaces.

- We write $X \sim Y$ if X and Y are *isomorphic*, i.e. there exists a linear bijection $T \colon X \to Y$ such that T and T^{-1} are continuous.
- We write $X \cong Y$ if X, Y are *isometrically isomorphic*, i.e. there exists a linear surjective map $T \colon X \to Y$ such that, for every $x \in X$, $||Tx|| = ||x||$.
- For $x \in X$ and $f \in X^*$, we write $\langle x, f \rangle$ for the *action of f on x*, i.e. $f(x)$.

Definition Positive homogeneous and subadditive

Let X be a real vector space. We say that a functional $p \colon X \to \mathbb{R}$ is

- *positive homogeneous if $p(tx) = tp(x)$ for all $x \in X$ and $t \geq 0$*
- *subadditive if $p(x + y) \leq p(x) + p(y)$ for all $x, y \in X$*

Theorem 1.1 Hahn–Banach I

Let X be a real vector space and $p \colon X \to \mathbb{R}$ a positive homogeneous and subadditive functional. Let $Y \subset X$ be a subspace and $g \colon Y \to \mathbb{R}$ a linear map such that $g(y) \leq p(y)$ for every $y \in Y$. Then there exists a linear map $f \colon X \to \mathbb{R}$ such that

(i) $f|_Y = g$

(ii) $f(x) \leq p(x)$ *for every* $x \in X$

Proof. Consider the set

$$\mathcal{P} := \left\{ (Z, h) \colon \begin{array}{l} Y \subset Z \subset X, Z \text{ subspace}, \\ h \colon Z \to \mathbb{R} \text{ linear}, \\ h|_Y = g, \\ h(z) \leq p(z) \ \forall z \in Z \end{array} \right\}$$

Note that $\mathcal{P}$ is partially ordered by extension, i.e. $(Z_1, h_1) \preceq (Z_2, h_2)$ iff $Z_1 \subset Z_2$ and $h_2|_{Z_1} = h_1$. By Zorn's lemma, $\mathcal{P}$ has a maximal element (Z, h).

Suppose, on the contrary, that $Z \neq X$. Then we can fix $x \in X \backslash Z$ and $\alpha \in \mathbb{R}$. Define $h_1 \colon Z_1 := Z + \mathbb{R}x \to \mathbb{R}$ by $h_1(z + \lambda x) = h(z) + \lambda\alpha$. By construction, h_1 is linear and $h_1|_Z = h$.

It then suffices to find α such that $h_1(z+\lambda x) \leq p(z+\lambda x)$ for every $z \in Z$ and $\lambda \in \mathbb{R}$, since this would imply that $(Z_1, h_1) \in \mathcal{P}$ — a contradiction! To this end, note that, since p is positive homogeneous, we need

$$h(z) + \alpha \leq p(z+x)$$
$$h(z) - \alpha \leq p(z-x)$$

for all $z \in Z$, i.e. for every $y, z \in Z$,

$$h(y) - p(y-x) \leq \alpha \leq p(z+x) - h(z)$$

Of course, it suffices to show that, for every $y, z \in Z$,

$$h(y) - p(y-x) \leq p(z+x) - h(z)$$

Indeed, by hypothesis, we have

$$h(y) + h(z) = h(y+z) \leq p(y+z) \leq p(y-x+x+z) \leq p(y-x) + p(x+z)$$

as required. ■

Definition Seminorm

Let X be a real or complex normed space. A seminorm on X is a function $p\colon X \to \mathbb{R}$ such that

(i) $p(x) \geq 0$ *for all* $x \in X$

(ii) $p(\lambda x) = |\lambda| p(x)$ *for all* $x \in X$ *and scalars* λ

(iii) $p(x+y) \leq p(x) + p(y)$ *for all* $x, y \in X$

Remark. A norm is a seminorm, which is in turn a positive homogeneous and subadditive functional.

Theorem 1.2 Hahn–Banach II

Let X be a real or complex normed space and p a seminorm on X. Let $Y \subset X$ be a subspace and g a linear functional on Y such that $|g(y)| \leq p(y)$ for all $y \in Y$. Then there exists a linear functional f on X such that $f|_Y = g$ and $|f(x)| \leq p(x)$ for all $x \in X$.

Proof. We shall first prove the real case and then extend it to the complex case.

Case 1: Real case

Note that p is a positive homogeneous and subadditive functional satisfying $g(y) \leq |g(y)| \leq p(y)$ for all $y \in Y$. By Theorem 1.1, there exists a linear functional $f\colon X \to \mathbb{R}$ such that $f|_Y = g$ and $f(x) \leq p(x)$ for all $x \in X$. For $x \in X$, $-f(x) = f(-x) \leq p(-x) = p(x)$, so $|f(x)| \leq p(x)$.

Case 2: Complex case

Let $g_1 = \operatorname{Re}(g)$. Then $g_1\colon Y \to \mathbb{R}$ is $\mathbb{R}$-linear and, for every $y \in Y$, $|g_1(y)| \leq |g(y)|$. By the real case, there exists an $\mathbb{R}$-linear $f_1\colon X \to \mathbb{R}$ such that $f_1|_Y = g_1$ and, for every $x \in X$, $|f_1(x)| \leq p(x)$.

▶ **Claim.** There exists a unique linear map $f\colon X \to \mathbb{C}$ such that $\operatorname{Re} f = f_1$.

Proof. We simply take $f(x) = f_1(x) - i f_1(ix)$.

Note that $\operatorname{Re}(f|_Y) = f_1|_Y = g_1 = \operatorname{Re}(g)$, so $f|_Y = g$ by uniqueness. For $x \in X$, we have $|f(x)| = \lambda f(x) = f(\lambda x) = f_1(\lambda x) \leq p(\lambda x) = |\lambda| p(x)$ for some $\lambda \in \mathbb{C}, |\lambda| = 1$. ■

Remark. For a complex vector space V, we shall denote by $V_{\mathbb{R}}$ for V viewed as a real vector space. Then the proof of Theorem 1.2 shows that, for a complex normed space X, the map

$$\begin{aligned}(X^*)_{\mathbb{R}} &\to (X_{\mathbb{R}})^* \\ f &\mapsto \operatorname{Re} f\end{aligned}$$

is an isometric isomorphism.

Corollary 1.3
Let X be a real or complex vector space and p a seminorm on X. For any $x_0 \in X$, there exists a linear functional f on X such that

(i) *for every $x \in X$, $|f(x)| \leq p(x)$*
(ii) $f(x_0) = p(x_0)$

Proof. Let $Y = \operatorname{span}\{x_0\}$. Define a linear functional g on Y by setting $g(\lambda x_0) = \lambda p(x_0)$. Then for every scalar λ, $|g(\lambda x_0)| = |\lambda| p(x_0) = p(\lambda x_0)$. By Theorem 1.2, there exists a linear functional f on X such that $f|_Y = g$ and, for every $x \in X$, $|f(x)| \leq p(x)$. Of course, we also have $f(x_0) = g(x_0) = p(x_0)$. ∎

Theorem 1.4 Hahn–Banach III
Let X be a real or complex normed space.

(i) *Given a subspace Y of X and $g \in Y^*$, there exists $f \in X^*$ such that $f|_Y = g$ and $||f|| = ||g||$.*
(ii) *Given $x_0 \in X \backslash \{0\}$, there exists $f \in S_{X^*}$ such that $f(x_0) = ||x_0||$.*

Proof. To prove (i), we define $p\colon X \to \mathbb{R}$ by $p(x) := ||g||\,||x||$. This is a seminorm on X and, for every $y \in Y$, $|g(y)| \leq ||g||\,||y|| = p(y)$. We then proceed by applying Theorem 1.2 to obtain some $f \in X^*$ with $f|_Y = g$ and $||f|| \leq ||g||$. Since $f|_Y = g$, we immediately have $||f|| \geq ||g||$. For (ii), we take $p = ||\cdot||$. Apply the previous corollary. ∎

Remarks

- Theorem 1.4(i) is a linear version of Tietze's extension theorem (cf. II Linear Analysis).
- Theorem 1.4(ii) implies that X^* separates points of X, i.e. given $x \neq y$ in X, there exists $f \in X^*$ such that $f(x) \neq f(y)$. (Indeed, simply take $x_0 = x - y$.)
- The f in Theorem 1.4(ii) is sometimes called a *norming functional at* x_0:

  $$||x_0|| = \max\{|\langle x_0, g\rangle|\colon g \in B_{X^*}\}$$

  Alternatively, f is also called a *support functional at* x_0 (cf. separation theorems in II Analysis of Functions).

1.2 Biduals

Let X be a normed space. We call $X^{**} := (X^*)^*$ the *bidual* or *second dual* of X. Recall from II Linear Analysis that it is a Banach space in the operator norm. For each $x \in X$, define the map

$$\begin{aligned}\hat{x}\colon X^* &\to \mathbb{R} \\ f &\mapsto f(x)\end{aligned}$$

Note that $\hat{x}$ is linear and bounded: $|\hat{x}(f)| = |f(x)| \leq ||f||\,||x||$. So $\hat{x} \in X^{**}$ with $||\hat{x}|| \leq ||x||$. Thus, we can then define the *canonical embedding* of X into X^{**}

$$\hat{}\colon X \to X^{**}$$

$$x \mapsto \hat{x}$$

Theorem 1.5
The map $\hat{}$ is an isometric isomorphism of X into X^{**}.

Proof. The map is well-defined and norm-decreasing by above. It is easy to check linearity. The map is isometric since

$$||\hat{x}|| = \sup\{|\hat{x}(f)| \colon f \in B_{X^*}\} = ||x||$$

by Theorem 1.4(ii). ∎

Remarks

- For $x \in X$ and $f \in X^*$, we have $\langle f, \hat{x} \rangle = \langle x, f \rangle$.
- $X \cong \hat{X} := \hat{}(X) \subset X^{**}$, so $\hat{X}$ is closed in X^{**} iff X is complete.
- In general, $\overline{\hat{X}}$ (closure in X^{**}) is a Banach space containing an isometric copy of X as a dense subspace.

Definition Reflexive
We say that X is reflexive if $\hat{X} = X^{**}$.

Examples

- Reflexive spaces: ℓ_p for any $p \in (1, \infty)$, any Hilbert space, any finite-dimensional space (see also $L_p(\mu)$ later on)
- Not reflexive spaces: $c_0, \ell_1, L_1[0,1]$

Remark. If X is reflexive, then $X \cong X^{**}$. But there exist Banach spaces X such that $X \cong X^{**}$ but X is not reflexive.

1.3 Dual operators

Let X, Y be normed spaces. We denote by $\mathcal{B}(X,Y) := \{T \colon X \to Y \mid T \text{ is linear and bdd}\}$. Recall from II Linear Analysis that this is a normed space in the operator norm $||T|| := \sup\{||Tx|| \colon x \in B_X\}$ and that $\mathcal{B}(X,Y)$ is complete if Y is complete.

For $T \in \mathcal{B}(X,Y)$, the *dual operator* of T is the map

$$T^* \colon Y^* \to X^*$$
$$g \mapsto g \circ T$$

It is easy to see that T is a well-defined linear map and it is bounded with $||T^*|| = ||T||$:

$$||T^*|| = \sup_{g \in B_{Y^*}} ||T^* g|| = \sup_{g \in B_{Y^*}} \sup_{x \in B_X} |\langle x, T^*(g) \rangle| = \sup_{x \in B_X} \sup_{g \in B_{Y^*}} |\langle Tx, g \rangle| = \sup_{x \in B_X} ||Tx|| = ||T||$$

where the penultimate equality follows from Theorem 1.4(ii).

Remark. When X, Y are Hilbert spaces, the dual operator T^* 'is' the adjoint of T.

Example. For $1 < p, q < \infty$ with $p^{-1} + q^{-1} = 1$, the dual of the right-shift operator $R \colon \ell_p \to \ell_p$ is simply the left-shift operator $L \colon \ell_q \to \ell_q$.

Now, let's note some properties of dual operators:

(1) $(\mathrm{Id}_X)^* = \mathrm{Id}_{X^*}$
(2) $(\lambda S + \mu T)^* = \lambda S^* + \mu T^*$ for all $S, T \in \mathcal{B}(X,Y)$ and all scalars λ, μ

(3) $(ST)^* = T^*S^*$ for all $T \in \mathcal{B}(X,Y)$ and all $S \in \mathcal{B}(Y,Z)$

(4) The map

$$\begin{aligned} \mathcal{B}(X,Y) &\to \mathcal{B}(Y^*, X^*) \\ T &\mapsto T^* \end{aligned}$$

is an into isometric isomorphism.

(5) Let $T \in \mathcal{B}(X,Y)$. The diagram

$$\begin{array}{ccc} X & \xrightarrow{T} & Y \\ {\scriptstyle\wedge}\downarrow & & \downarrow{\scriptstyle\wedge} \\ X^{**} & \xrightarrow[T^{**}]{} & Y^{**} \end{array}$$

commutes, i.e., $\widehat{Tx} = T^{**}\hat{x}$ for all $x \in X$. For $x \in X, g \in Y^*$,

$$\langle g, T^{**}\hat{x}\rangle = \langle T^*g, \hat{x}\rangle = \langle x, T^*g\rangle = \langle Tx, g\rangle = \langle g, \widehat{Tx}\rangle$$

Remark. Properties (1) and (3) imply that $X \sim Y \implies X^* \sim Y^*$.

1.4 Quotient spaces

Let X be a normed space and Y a closed subspace. Then X/Y is a normed space in the quotient norm defined by

$$||x + Y|| := d(X,Y) := \inf\{||x+y|| \colon y \in Y\}$$

The quotient map $q\colon X \to X/Y$ is linear, surjective and bounded with $||q|| = 1$. Indeed, we have $||q(x)|| \le ||x||$ so $||q|| \le 1$. Note also that $q(D_X) = D_{X/Y}$ (for $x \in D_X$, we have $||q(x)|| \le ||x|| < 1$, so $q(x) \in D_{X/Y}$) and that, given $z \in D_{X/Y}$, we can pick some $x \in X$ such that $q(x) = z$ and, since $||z|| < 1$, we have $||x + y|| < 1$ for some $y \in Y$ and thus $x + y \in D_X$ with $q(x+y) = z$. It then follows that $||q|| = 1$ and q is an open map.

Given another normed space Z and a bounded linear map $T\colon X \to Z$ such that $Y \subset \ker T$, there exists a unique map $\tilde{T}$ such that

$$\begin{array}{ccc} X & \xrightarrow{\quad T \quad} & Z \\ & {\scriptstyle q}\searrow \quad \nearrow{\scriptstyle \tilde{T}} & \\ & X/Y & \end{array}$$

commutes, i.e. $T = \tilde{T} \circ q$. Moreover, $\tilde{T}$ is linear and bounded with $||\tilde{T}|| = ||T||$ since

$$\tilde{T}(D_{X/Y}) = \tilde{T}(q(D_X)) = T(D_X)$$

Theorem 1.6
Let X be a normed space. If X^* is separable, then so is X.

Proof. Pick a countable set $\{f_n \colon n \in \mathbb{N}\}$ dense in S_{X^*}. For every $n \in \mathbb{N}$, we can pick some $x_n \in B_X$ such that $f_n(x_n) > \frac{1}{2}$. Let $Y = \overline{\operatorname{span}}\{x_n \colon n \in \mathbb{N}\}$. It suffices to show that $Y = X$. Suppose, on the contrary, that $Y \neq X$. Then we can pick $g \in S_{X^*}$ such that $g|_Y = 0$. This is possible since we can, for instance, pick some $h \in (X/Y)^*$ such that $h \neq 0$ [e.g. Theorem 1.4(ii)], and we can then take $g = (h \circ q)/||h \circ q||$. Then there exists some $n \in \mathbb{N}$ such that $||f_n - g|| < \frac{1}{10}$. Then $||g(x_n)|| \ge \frac{1}{2} - \frac{1}{10} > 0$, which is a contradiction. ∎

Remark. The converse is false in general. For instance, the dual of $X = \ell_1$ is $X^* = \ell_\infty$, which is not separable.

Theorem 1.7
Any separable normed space X embeds isometrically into ℓ_∞.

Proof. Let $\{x_n : n \in \mathbb{N}\}$ be dense in X. For each $n \in \mathbb{N}$, let f_n be a norming functional at x_n, cf. Theorem 1.4(ii). We then define $T : X \to \ell_\infty$ by $Tx = (f_n(x))_{n=1}^\infty$. This is a bounded linear map since, for every $n \in \mathbb{N}$, we have $|f_n(x)| \leq ||x||$, so $Tx \in \ell_\infty$ and $||Tx||_\infty \leq ||x||$. For every $n \in \mathbb{N}$, we also have $||Tx_n||_\infty \geq |f_n(x_n)| = ||x_n||$ and thus $||Tx_n||_\infty = ||x_n||$. Hence, by density and continuity, it follows that $||Tx||_\infty = ||x||$. ∎

Remarks

- ℓ_∞ is isometrically universal for the class $\mathcal{SB}$ of separable Banach spaces. In fact, we will see later that there exists $Z \in \mathcal{SB}$ isometrically universal for $\mathcal{SB}$.
- A dual version: every separable Banach space is a quotient of ℓ_1.

Theorem 1.8 Vector-valued Liouville
Let X be a complex Banach space and $f : \mathbb{C} \to X$ holomorphic and bounded. Then f is constant.

Proof. Pick some $M > 0$ such that $||f(z)|| \leq M$ for all $z \in \mathbb{C}$. Since f is holomorphic, we have that, for every $z \in \mathbb{C}$, the limit

$$\lim_{w \to z} \frac{f(w) - f(z)}{w - z}$$

exists and shall be denoted by $f'(z)$. Now, fix $\varphi \in X^*$. Then $\varphi \circ f : \mathbb{C} \to \mathbb{C}$ is bounded

$$|\varphi(f(z))| \leq ||\varphi||||f(z)|| \leq M||\varphi||$$

and holomorphic

$$\lim_{w \to z} \frac{\varphi(f(w)) - \varphi(f(z))}{w - z} = \varphi\left(\lim_{w \to z} \frac{f(w) - f(z)}{w - z}\right)$$

where we have used the fact that φ is linear and continuous. By scalar Liouville, we have that $\varphi(f(z)) = \varphi(f(0))$ for every $z \in \mathbb{C}$.

Fix $z \in \mathbb{C}$. By above, we have that, for every $\varphi \in X^*$, $\varphi(f(z)) = \varphi(f(0))$. By Theorem 1.4(ii), it follows that $f(z) = f(0)$. ∎

1.5 Locally convex spaces

Definition Locally convex space (LCS)
A locally convex space is a pair $(X, \mathcal{P})$, where X is a (real or complex) vector space and $\mathcal{P}$ is a family of seminorms on X that separates points of X in the sense that, for every nonzero $x \in X$, there exists $p \in \mathcal{P}$ such that $p(x) \neq 0$.

Note that $\mathcal{P}$ induces a topology on X: $U \subset X$ is open iff

$$\forall\, x \in U\ \exists\, n \in \mathbb{N}\ \exists\, p_1, \cdots, p_n \in \mathcal{P}\ \exists\, \varepsilon > 0 \text{ s.t. } \{y \in X : p_k(y - x) < \varepsilon, 1 \leq k \leq n\} \subset U$$

Some basic properties of this topology include

- Addition and scalar multiplication are continuous
- It is Hausdorff
- $x_n \to x$ iff $p(x_n - x) \to 0$ for all $p \in \mathcal{P}$

- Let Y be a subspace of X. Let $\mathcal{P}_Y := \{p|_Y : p \in \mathcal{P}\}$. Then $(Y, \mathcal{P}_Y)$ is a LCS and its topology is the subspace topology induced by X on Y.
- Given two families $\mathcal{P}, \mathcal{Q}$ on X, they are equivalent (denoted by $\mathcal{P} \sim \mathcal{Q}$) if they induce the same topology on X. The topology of $(X, \mathcal{P})$ is metrisable iff there exists a countable $\mathcal{Q} \sim \mathcal{P}$.

Definition Fréchet space
A Fréchet space is a complete metrisable LCS.

Examples

- Any normed space $(X, ||\cdot||)$ is a LCS with $\mathcal{P} = \{||\cdot||\}$.
- Fix a nonempty open $U \subset \mathbb{C}$. Define

$$\mathcal{O}(U) := \{f : U \to \mathbb{C} \mid f \text{ is holomorphic}\}$$

  For a compact $K \subset U$, let $p_K(f) := \sup_{z \in K} |f(z)|$ for every $f \in \mathcal{O}(U)$. Let $\mathcal{P} = \{p_K : \text{ compact } K \subset U\}$. Then $(\mathcal{O}(U), \mathcal{P})$ is a LCS and its topology is the topology of local uniform convergence. It is also a Fréchet space and not normable, i.e. the topology is not induced by a norm.
- Fix $d \in \mathbb{N}$ and a nonempty open $\Omega \subset \mathbb{R}^d$. We define

$$C^\infty(\Omega) := \{f : \Omega \to \mathbb{R} \mid f \text{ is smooth}\}$$

  For a multi-index $\alpha \in \mathbb{Z}^d_{\geq 0}$, we can define

$$D^\alpha := \left(\frac{\partial}{\partial x_1}\right)^{\alpha_1} \cdots \left(\frac{\partial}{\partial x_d}\right)^{\alpha_d}$$

  For compact $K \subset \Omega$, we define

$$p_{K,\alpha}(f) := \sup_{w \in K} |(D^\alpha f)(w)|$$

  We then take $\mathcal{P} = \{p_{K,\alpha} : \text{ compact } K \subset \Omega, \alpha \in \mathbb{Z}^d_{\geq 0}\}$. Then $(C^\infty(\Omega), \mathcal{P})$ is a LCS. It is also Fréchet space and not normable.